

- Spending flexibility: is it possible to reduce portfolio distributions without harming your standard of living in a significant way?
- Availability of reserves: what other resources are available that have not been earmarked to manage spending and can be used to cover contingencies?

These factors all relate to what is an acceptable probability of success, or probability of failure, for the retirement plan. For someone who worries and loses sleep about outliving his or her portfolio, does not have much additional income from outside the portfolio, mostly faces fixed expenses without much room to make cuts, and does not have much in the way of backup reserves, it may be necessary to plan for a quite high probability of success. This will imply using a lower stock allocation and a lower spending rate. However, for someone who has less fear about outliving his or her portfolio, has a number of additional income sources from outside the portfolio, has the flexibility to cut portfolio spending without adversely impacting lifestyle, and has sufficient additional reserves, a higher spending rate accompanied by a lower probability of success could be quite satisfactory and “optimal.”

The first to explore this were Moshe Milevsky and Huaxiong Huang in their March/April 2011 *Financial Analysts Journal* article titled “Spending Retirement on Planet Vulcan: The Impact of Longevity Risk Aversion on Optimal Withdrawal Rates.” By referring to Planet Vulcan from Star Trek, they mean to discuss the decisions made by a fully rational investor when facing longevity risk but not market risk. The individual does not know how long he or she will live, but the investment returns are known in advance. The retiree has to decide how much to spend from the portfolio each year in the face of an uncertain lifespan. The authors summarize the rational investor’s decision-making process as follows: “Wealth managers should advocate dynamic spending in proportion to survival probabilities, adjusted up for exogenous pension income and down for longevity risk aversion.”

Several important points are packed into that sentence. The first idea is that retirees should intentionally plan to spend more when they are sure to be alive, while planning for reduced spending later on when survival is less